

K. J. Somaiya College of Engineering, Mumbai-77

End Semester Exam

December 2020

Max. Marks: 30

Duration: 1 Hr. 15 Min. (For attempting questions) + 15 min. (uploading)

Class: SY/B.Tech

Semester: III

Name of the Course: Data Communication and Networking

Branch: IT

Course Code: 2UIC305

Instructions:

- (1) **All questions are compulsory**
- (2) **Draw neat diagrams**
- (3) **Assume suitable data if necessary**

Question No.		Max. Marks
Q.1	a) Explain any five factors that determine whether a communication system is LAN or WAN. b) Describe what are the three domains of the DNS ? OR a) Describe Client server paradigm with neat diagram ? b) Explain any five networking devices and also write the name of layer where they are useful.	10
Q.2	Attempt any two(05 M Each):- a) A block of addresses is granted to a small organization. One of the address is 205.16.37.40/28. What is the first address, last address and the number of addresses in the block . For the same IP in classful addressing scheme identify the class and network address. b) Differentiate between Datagram Network and Virtual-Circuit Network w.r.t i) Circuit setup ii) Addressing iii) State information iv) Routing and v) Effect of router failures. c) Explain any two guided and three unguided medium with their applications.	10
Q.3	A Slotted ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces a) 1000 frames per second b) 500 frames per second c) 250 frames per second	10